

CampusEats — External SOAP Partner Integration

Team ID: 13

Team Members:

1. Aman Jha Roll No. 20251651016
2. Himanshu singh deopa Roll No. 20251651044
3. Md.Kaif Roll No. 20251651055

Task 1 — Context

Context: CampusEats Payment Gateway Integration

CampusEats uses REST APIs for most of its internal and student-facing services because REST is lightweight and easy to integrate. However, the external payment gateway will use SOAP because payment processing requires a strong and well-defined contract between CampusEats and the partner. SOAP also supports message-level security and structured fault handling, which are useful for sensitive payment operations. The integration will use the `charge` operation to send payment details and receive either a successful transaction response or a payment-related fault.

Task 4 — HTTP Binding

HTTP Request

<http>

POST /soap/payment HTTP/1.1

Host: api.campuspay.example.com

Content-Type: text/xml; charset=utf-8

SOAPAction: <http://campuseats.example.com/payment/charge>

<?xml version="1.0" encoding="UTF-8"?>

<soap:Envelope

xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"

xmlns:pay="http://campuseats.example.com/payment">

<soap:Header>

<pay:Credentials>

<pay:merchantId>CEATS001</pay:merchantId>

<pay:apiKey>CampusEatsSecureKey</pay:apiKey>

</pay:Credentials>

</soap:Header>

<soap:Body>

<pay:chargeRequest>

<pay:orderId>ORD10045</pay:orderId>

```

        <pay:amount>450.00</pay:amount>

        <pay:currency>INR</pay:currency>

        <pay:cardToken>tok_78451236</pay:cardToken>

    </pay:chargeRequest>

</soap:Body>

</soap:Envelope>

```

Task 5 Discovery

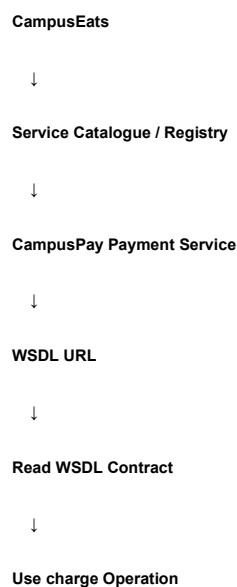
Service Discovery

CampusEats would obtain the partner's WSDL through a service catalogue/registry. The catalogue contains information about the external business, its SOAP service, service endpoint, and the WSDL that defines the service contract. CampusEats can look up the payment service in the catalogue and obtain the WSDL URL. The WSDL is then used by the CampusEats development team to understand the available operation, request/response structure, SOAP binding, and endpoint.

Registry / Catalogue Entry

Field	Details
Business	CampusPay Payment Services
Service	CampusPay SOAP Payment Service
Endpoint	https://api.campuspay.example.com/soap/payment
WSDL	https://api.campuspay.example.com/wSDL/payment.wSDL
Model-style Pointer	https://api.campuspay.example.com/wSDL/payment.wSDL
Operation	charge

Discovery Flow



Thus, CampusEats discovers the external SOAP partner through the catalogue and obtains the WSDL contract before implementing the integration.

6. Fault Mapping

The external payment partner may return a SOAP fault when a payment cannot be processed. For example, the partner may return the error:

CARD_DECLINED

This is the partner-specific vocabulary and should not be exposed directly to CampusEats students or consumers.

The CampusEats integration layer translates the external partner fault into the error defined by the CampusEats contract.

Fault Mapping

External Partner Fault	CampusEats Error
CARD_DECLINED	payment declined
Payment instrument was declined	Payment could not be completed

Mapping Process

CampusPay SOAP Fault

↓

CARD_DECLINED

↓

CampusEats Integration Layer

↓

payment declined

↓

CampusEats placeOrder Contract

For example, if the CampusPay gateway returns:

```
<pay:code>CARD_DECLINED</pay:code>
```

CampusEats converts this into its own application-level error:

payment declined

Therefore, the external partner's terminology does not leak into the CampusEats contract. Students and CampusEats clients only receive the error defined by the CampusEats system